

Week 01 Field Manual v1.0

Software Engineering Reboot 2026

Development Environment & Foundations

Purpose

This Field Manual is an operational guide for the 7:00 PM–8:00 PM study session. Its purpose is to eliminate decision fatigue and provide exact actions, commands, verification steps, evidence collection requirements, AI exercises, and journal prompts.

Engineering Context: Then vs Now

You spent the first eight years of your career doing software development and the next portion primarily in engineering leadership and management. Then: CVS, SVN, physical servers, waterfall, manual deployments. Now: Git, GitHub, Docker, cloud infrastructure, CI/CD, AI-assisted development. This week reconnects your prior engineering knowledge to the modern ecosystem.

Day 1 - Workstation Setup

7:00 Create root folders. Commands: `mkdir C:\Projects` `mkdir C:\Projects\Engineering-Reboot-2026`
7:10 Install Python. Verify: `python --version` 7:20 Install Git. Verify: `git --version` 7:30 Configure Git:
`git config --global user.name "Glenn Ogapong"` `git config --global user.email "glennj.ogapong@gmail.com"` 7:40 Install VS Code and extensions: Python, GitLens, Docker, Error Lens, Markdown All in One. 7:55 Journal: What worked? What failed? What surprised me?
Evidence: Screenshot Python version. Screenshot Git version. Screenshot VS Code extensions.

Day 2 - Git & GitHub

Goal: Create first repository and first commit. Commands: `cd C:\Projects\Engineering-Reboot-2026`
`mkdir engineering-reboot-2026` `cd engineering-reboot-2026` `git init` Create README.md `git add .` `git`
`commit -m "Initial commit"` Verify: `git log` AI Exercise: Ask ChatGPT: 'Explain Git to someone who
managed software teams but has not coded for 15 years.' Evidence: Repository screenshot.
Commit history screenshot.

Day 3 - Linux & WSL

Install WSL: `wsl --install` Verify: `wsl --status` Linux Lab: `pwd ls mkdir projects cd projects touch notes.txt ls` Challenge: Create folders projects, notes, archive using Linux commands only.
Evidence: Screenshot of WSL terminal.

Day 4 - Python Reawakening

Create hello.py print("Hello Glenn") Run: python hello.py Variables: name = "Glenn" age = 53 Input: name = input("Enter name: ") Mini Project 1: Age Calculator Mini Project 2: Temperature Converter Evidence: Screenshots of successful execution.

Day 5 - Calculator Project

Create calculator.py Functions: add subtract multiply divide Run: python calculator.py Publish: git add . git commit -m "Completed calculator project" git push Evidence: Working calculator. GitHub repository updated.

Command Reference

Git: git init git add . git commit -m "message" git push git pull git status git log Linux: pwd ls cd mkdir cp mv rm Python: python script.py

Troubleshooting

Python not found: Reinstall Python and enable Add To PATH. Git not found: Reinstall Git and restart terminal. WSL failure: Verify virtualization support. GitHub push failure: Verify authentication and repository permissions.

AI Companion Workflow

Every day: 1. Ask an AI assistant to explain the topic. 2. Validate against official documentation. 3. Record what was useful. 4. Record what was inaccurate or incomplete. Goal: Develop AI-assisted engineering judgment.

EB2-NIW Connection

Week 1 establishes the tooling foundation required to build future prototypes for the SME Digital Resilience & Economic Competitiveness Platform. Python will support analytics and automation. Git will support source control. Linux will support deployment. Docker will be introduced in future weeks.

Daily Journal Template

Date: Topic: What did I learn? What confused me? What did AI help me understand? What evidence did I collect? What will I do tomorrow?

End of Week Review

Can you: - Explain Git and GitHub? - Run Python programs? - Navigate Linux directories? - Create and commit a repository? - Explain why Docker matters? - Explain why Python dominates AI development? Score: Attendance /10 Execution /10 Learning /10 Confidence /10